ChainOps
SMART CONTRACT OPERATIONS

ChainOps

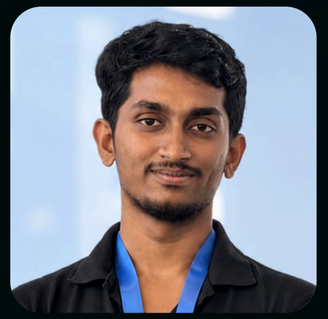
Smart Contract DevSecOps and
Operations Platform





IntelliCoders - Group 04
Faculty of Information Technology
University of Moratuwa

Meet Our Team



FERNANDO WWAST
244063P



JAYASIRI JMDS
244087R



RATHNASIRI LLKN
244176N



JAYAWICKRAMA JBAIN
244092D

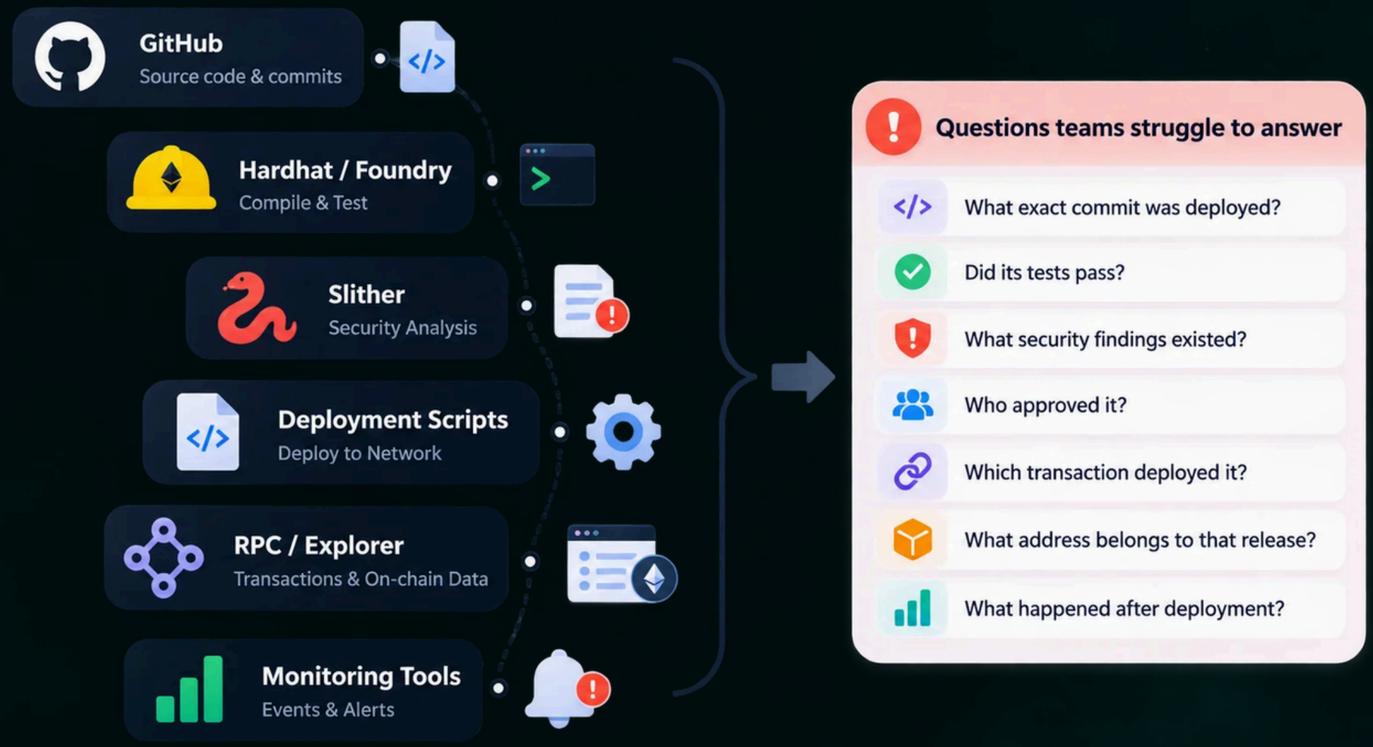


FAZREEN FF
244059G

PROJECT OVERVIEW

The Problem

Smart-contract delivery is fragmented across multiple tools



PROJECT OVERVIEW

Our Solution

ChainOps creates one traceable smart-contract lifecycle



ChainOps does not replace GitHub, Hardhat, Slither or blockchain infrastructure. It orchestrates them and adds lifecycle traceability, governance and operational visibility

AIM AND OBJECTIVE

Our Aim

Build a Smart Contract DevSecOps and operations platform connecting GitHub development with automated validation, security governance, controlled deployment and post-deployment monitoring.



AIM AND OBJECTIVE

Our Objectives

BUILD

- Automate source revision checkout, compilation and testing

SECURE

- Perform static analysis and evaluate release policies

GOVERN

- Control sensitive deployments through approval workflows

DEPLOY

- Track EVM transactions and maintain a contract registry

OBSERVE

- Monitor deployed contracts and generate operational alerts

SYSTEM SCOPE

What ChainOps includes and what is left out

IN SCOPE

- EVM-compatible Solidity projects
- GitHub integration
- Automated pipelines
- Build & testing
- Security analysis
- Policy & approvals
- Testnet deployment
- Contract registry
- Monitoring & alerts
- Audit & notifications

OUT OF SCOPE

- New blockchain
- New compiler
- New vulnerability analyzer
- Real-mainnet funds
- Every blockchain ecosystem
- Autonomous AI deployments
- Generic CI/CD for all software

TARGET USERS

Who uses and interacts with ChainOps



Solo development is also supported, where one developer can operate a project without requiring a formal organization structure.

KEY MODULES & FUNCTIONS

Six integrated areas powering the ChainOps lifecycle

Platform & Access
Auth • Projects • RBAC

Pipeline & Execution
GitHub • Build/Test • Runs

Security & Governance
Findings • Policies • Approvals

ChainOps
SMART CONTRACT OPERATIONS

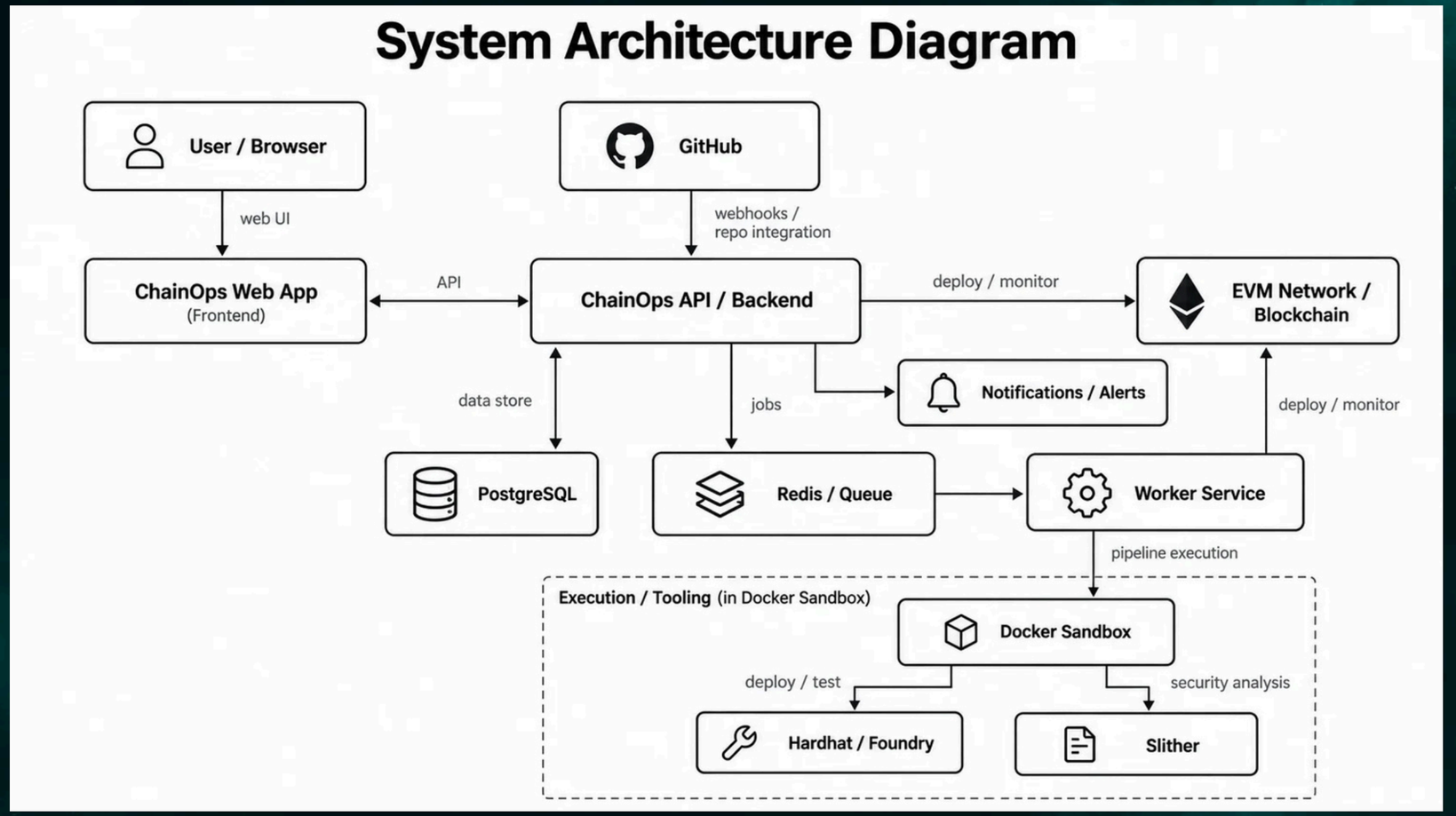
Deployment & Registry
EVM Deploy • Transactions • Registry

Monitoring & Alerts
Events • Rules • Notifications

Audit & AI
Audit Trail • Reports • Assistant

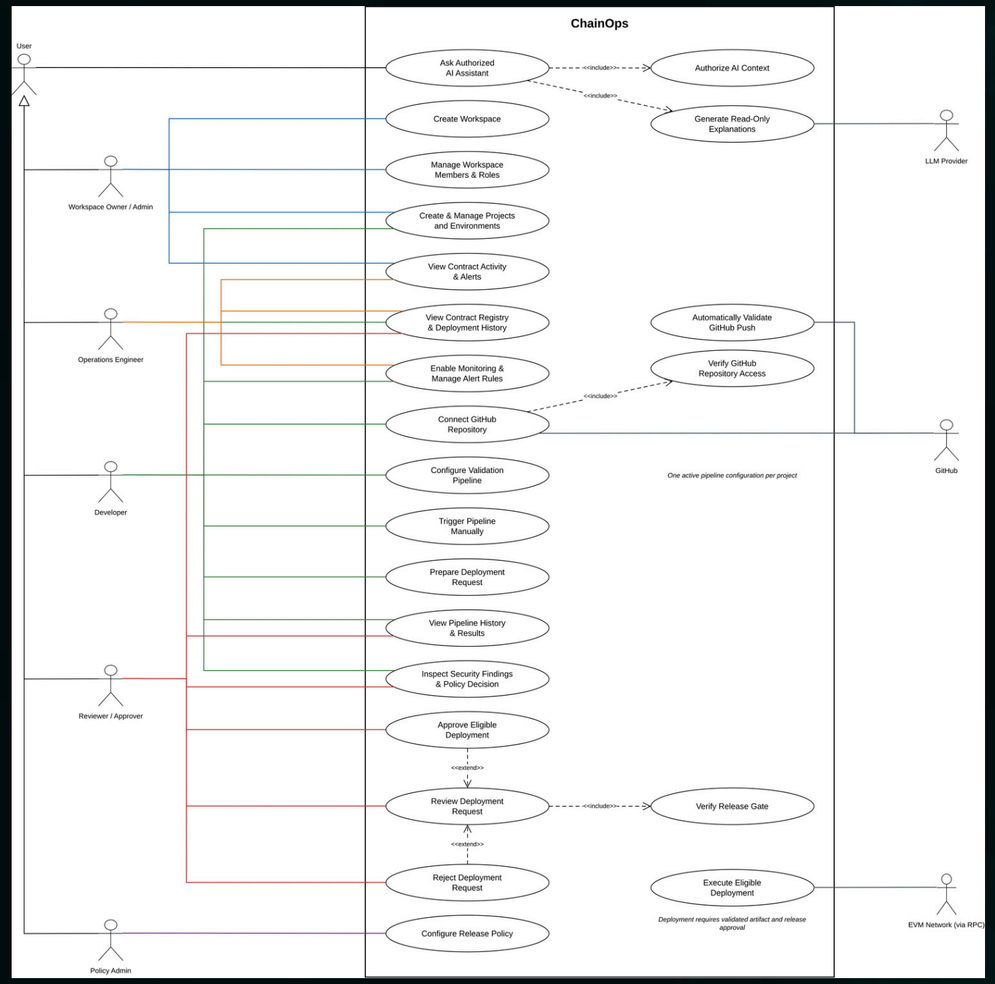
SYSTEM DIAGRAMS

System Architecture



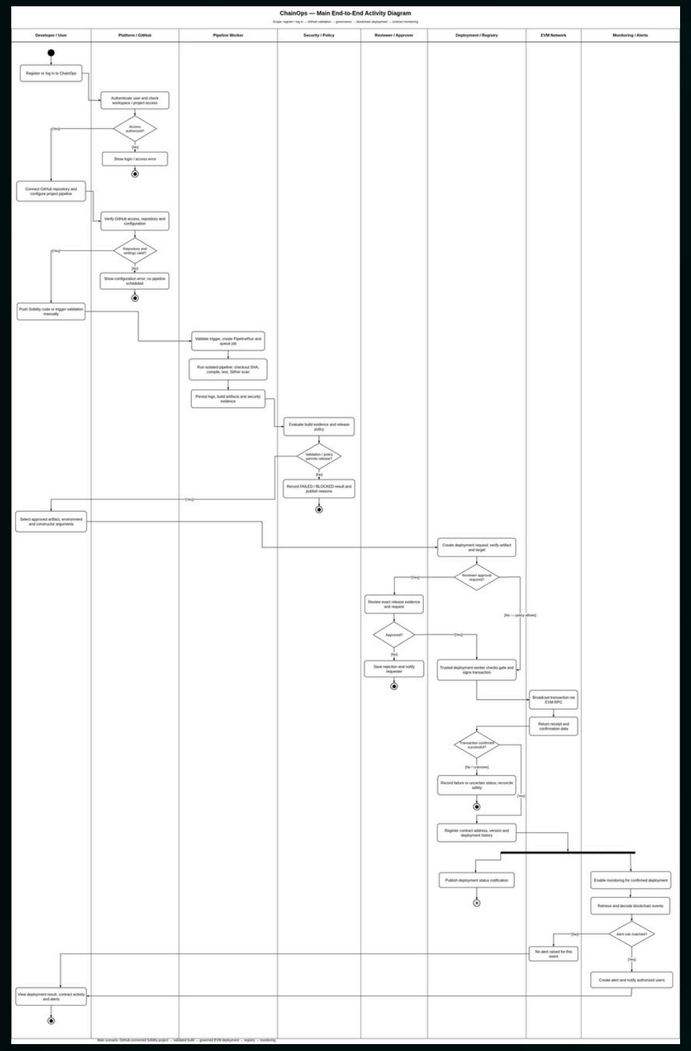
SYSTEM DIAGRAMS

Use-Case Diagram



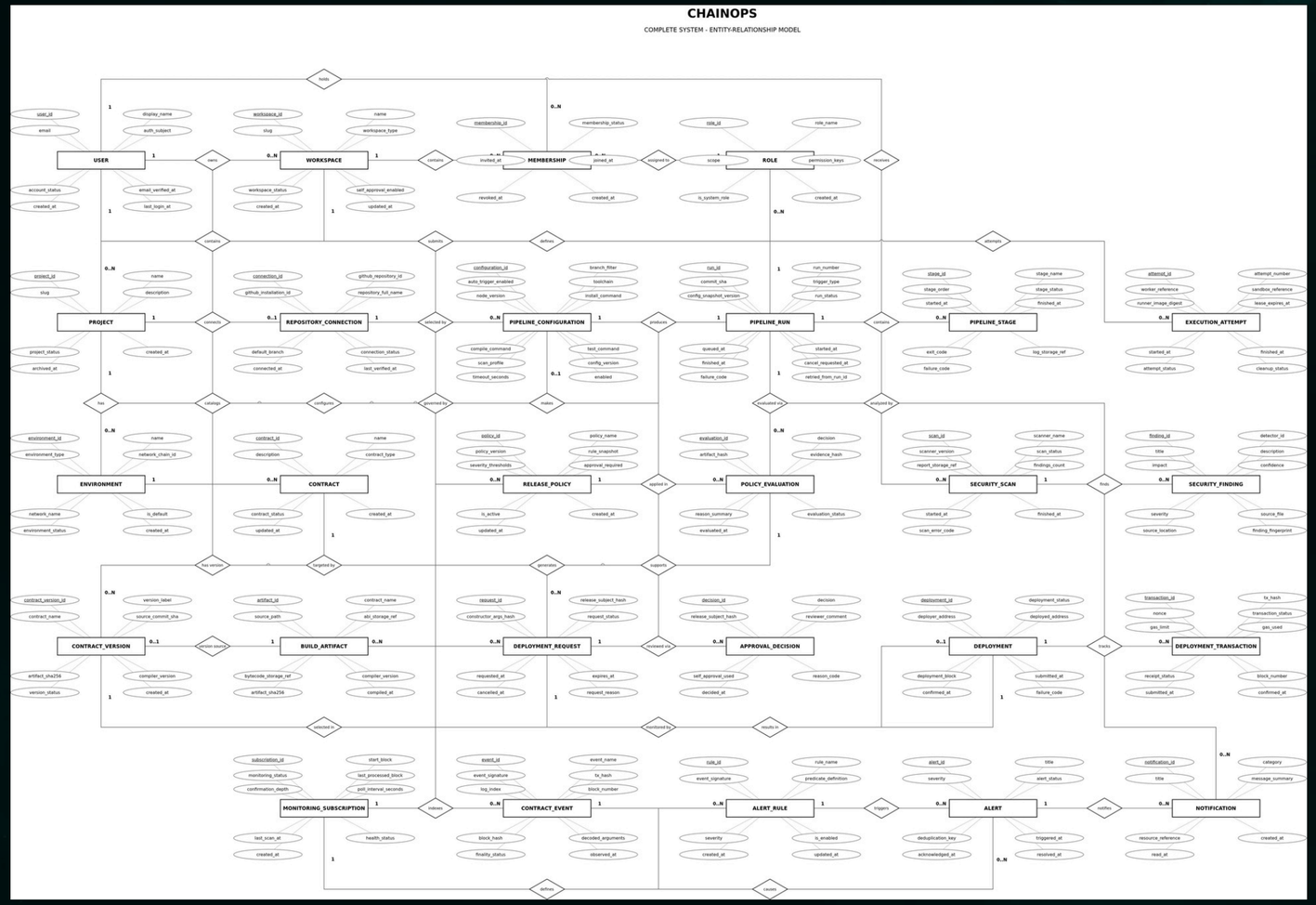
SYSTEM DIAGRAMS

Activity Diagram



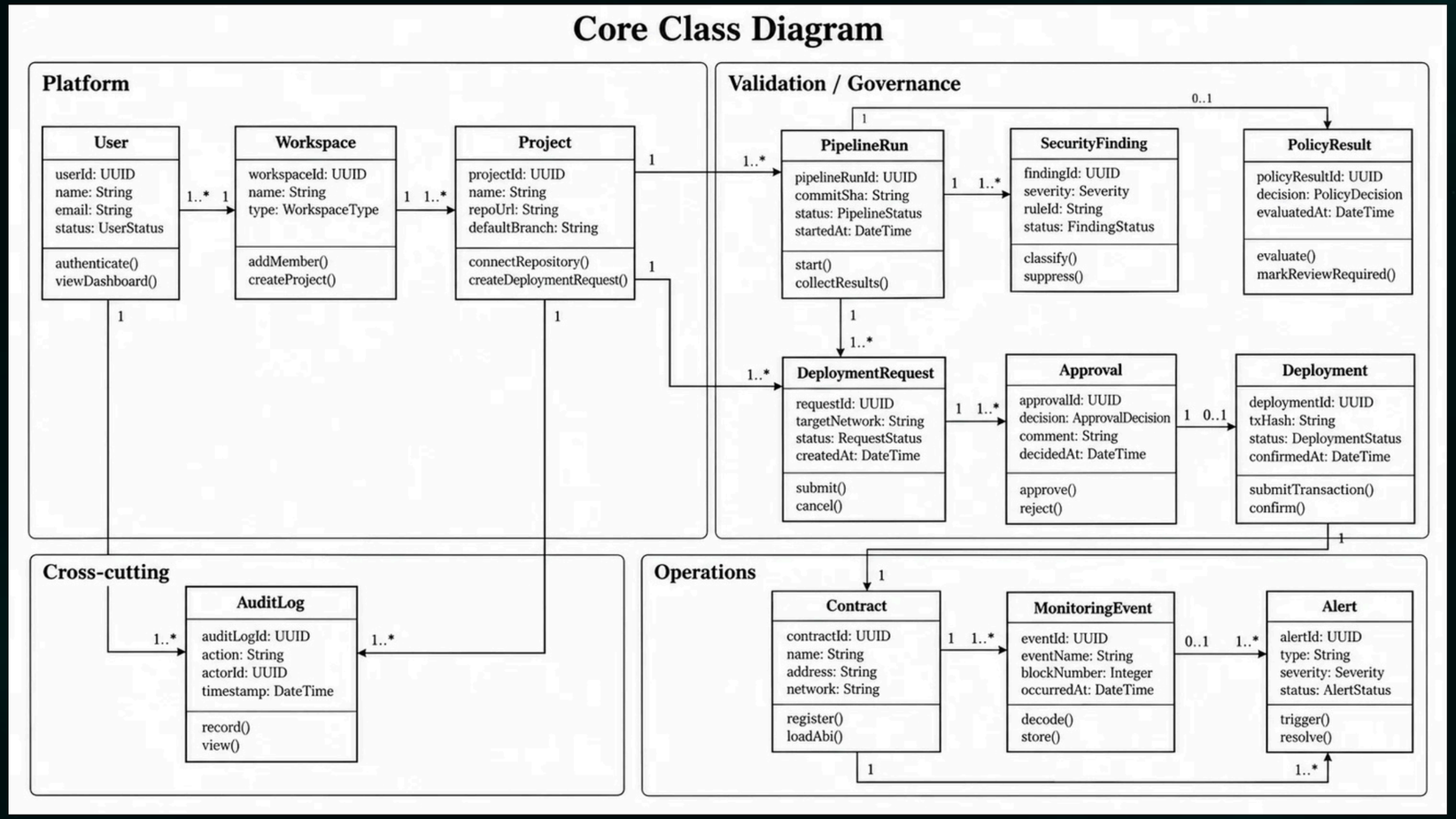
SYSTEM DIAGRAMS

ER Diagram



SYSTEM DIAGRAMS

Class Diagram



TECH STACK

Core Platforms and tools used in ChainOps

Frontend

- SvelteKit
- TypeScript

Backend & Data

- NestJS
- Prisma
- PostgreSQL
- Neon
- Redis
- BullMQ

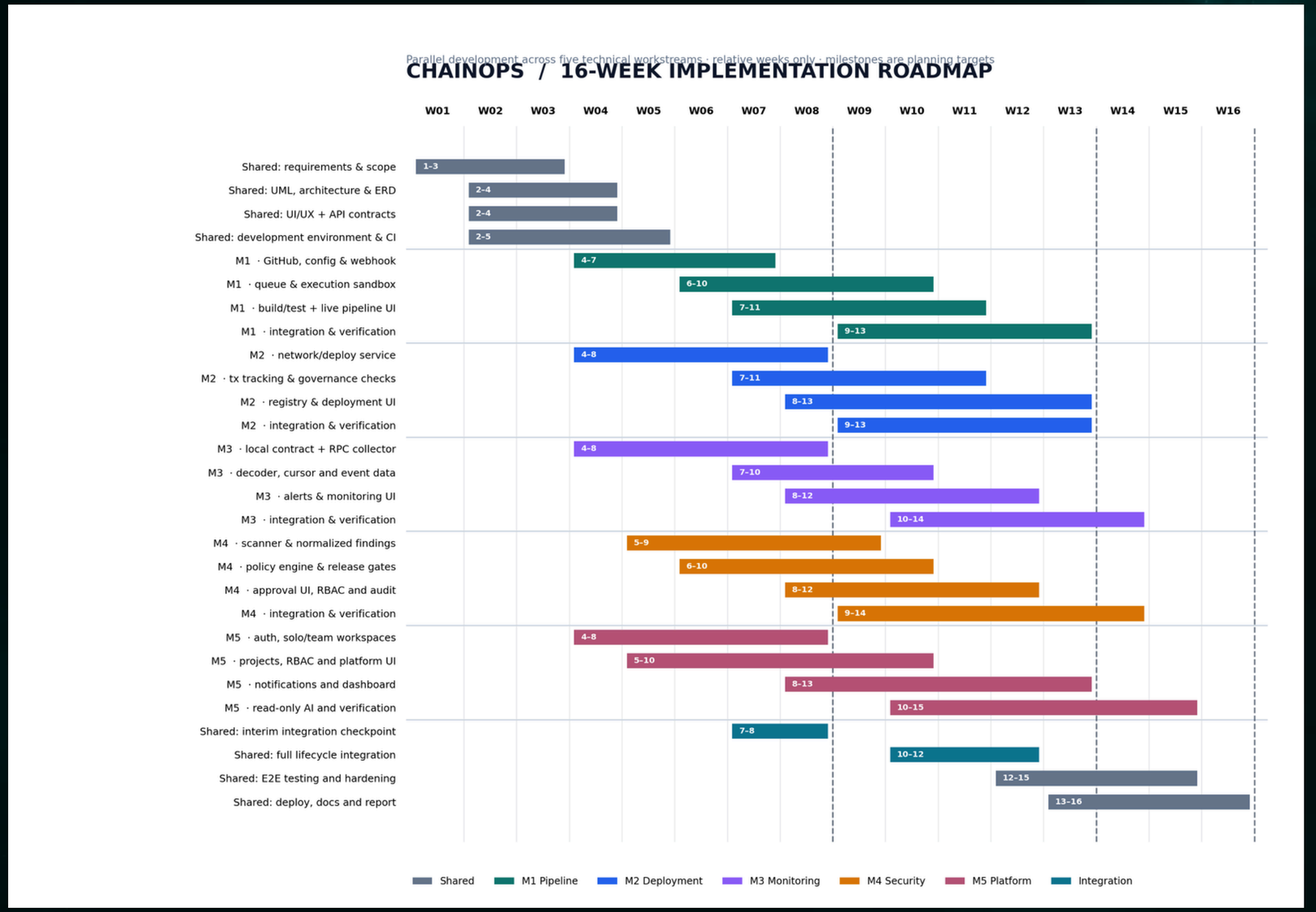
Blockchain & Security

- Solidity
- Hardhat
- viem
- Slither

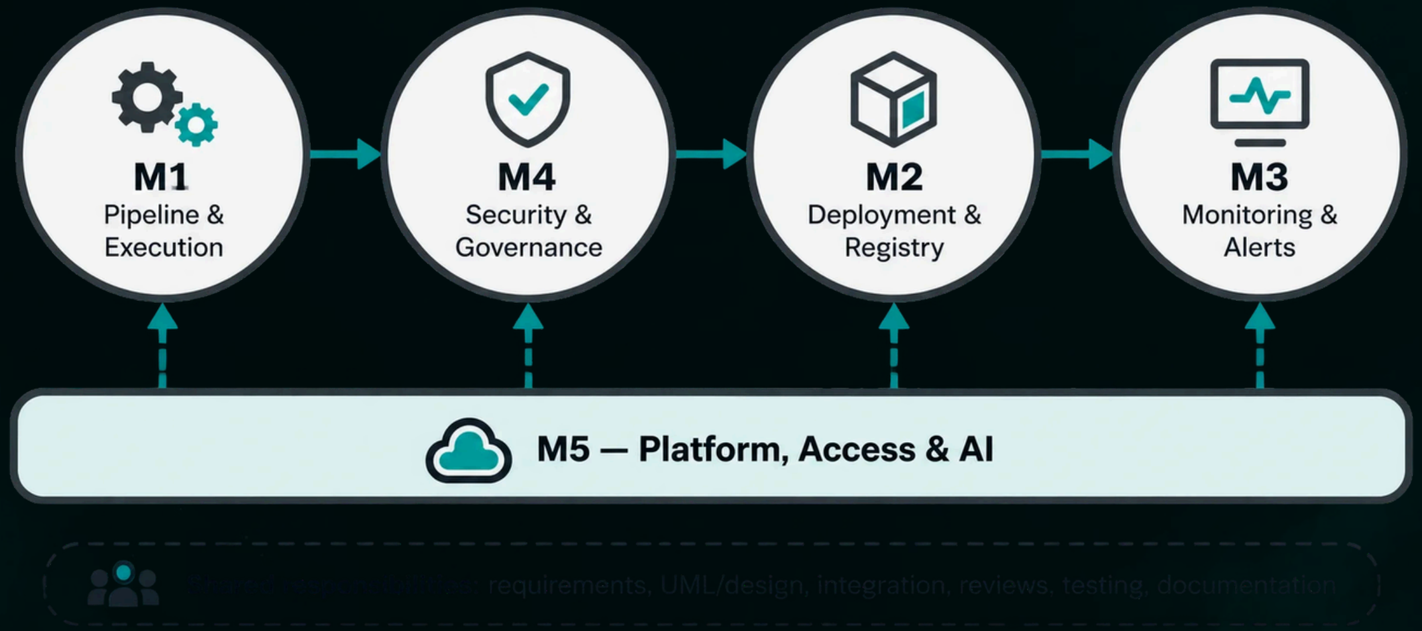
DevOps & Design

- Docker
- GitHub
- GitHub Actions
- Vercel
- Figma

PROJECT TIMELINE



INDIVIDUAL CONTRIBUTION



W.W.A.S.T FERNANDO

Pipeline & Execution

Frontend

- GitHub repository connection
- Pipeline setup and configuration
- Pipeline dashboard & run history
- Live pipeline status and execution logs
- Build and test result views

Backend

- GitHub webhook integration
- Pipeline run creation and lifecycle
- Queue & background worker processing
- Exact source-revision checkout
- Isolated build/test execution
- Compilation and automated testing
- Pipeline status, logs and failure handling

Key focus:
GitHub → Pipeline → Build/Test → Results

J.M.D.S. JAYASIRI

Blockchain Deployment & Contract Registry

Frontend

- Deployment request workflow
- Network and deployment configuration
- Live transaction status tracking
- Contract Registry
- Contract version and deployment history
- Contract details and deployed addresses

Backend

- Deployment validation and execution
- EVM / RPC integration
- Transaction signing and submission
- Confirmation and receipt tracking
- Deployment failure handling
- Contract registration
- Version and deployment-history management

Key focus:
Approved Release → Deployment → Transaction → Contract Registry

J.B.A.I.N. JAYAWICKRAMA

Monitoring, Events & Alerts

Frontend

- Contract monitoring dashboard
- Live contract activity feed
- Event and transaction history
- Alert-rule configuration
- Alert status and history
- Runtime analytics / monitoring views

Backend

- Blockchain event and log retrieval
- ABI-based event decoding
- Runtime event storage
- RPC / WebSocket monitoring
- Duplicate-event handling
- Alert-rule evaluation
- Alert creation and runtime-status processing

Key focus:
Deployed Contract → Events → Monitoring → Alerts

L.L.K.N. RATHNASIRI

Security & Governance

Frontend

- Security findings dashboard
- Vulnerability detail views
- Release-policy configuration
- Policy result / pass-block views
- Reviewer approval dashboard
- Approval / rejection workflow
- Governance audit history

Backend

- Security-analyzer integration
- Security finding normalization
- Release-policy evaluation
- Deployment eligibility gates
- Approval / rejection workflow
- Reviewer authorization checks
- Governance and audit records

Key focus:

Security Scan → Policy → Approval → Deployment Gate

F.F. FAZREEN

Platform, Access & AI Assistant

Frontend

- Login and registration
- Main ChainOps dashboard
- Workspace / organization management
- Project and environment management
- User and role management & Notification centre
- AI Assistant interface

Backend

- Authentication and session management
- Workspace / organization management
- Project and environment APIs
- Role-Based Access Control
- Shared authorization across modules
- Notification processing
- Restricted read-only AI Assistant
- Shared platform integration

Key focus:

User → Platform Access → RBAC → Shared Services → AI Assistant

CONCLUSION

ChainOps unifies smart contract development, security, deployment, and monitoring in one platform.
It simplifies complex workflows through automation, governance, and real-time visibility.
Our goal is to make smart contract operations more secure, efficient, and traceable.

Q & A

THANK YOU !

